

■ Quick Links

■ Telegram Bot (App)	https://t.me/Ops_agent_supermarket_bot
■ Telegram Bot (Web)	https://web.telegram.org/k/#@Ops_agent_supermarket_bot
■ GitHub Repository	https://github.com/lokesh2005-gif/Kirana-ops-agent
■ Live Server (Render)	https://kirana-ops-agent-dfh0.onrender.com

■ Project Overview

Kirana Ops Agent is an AI-powered Telegram bot designed for Indian grocery (kirana) store owners. It lets the shopkeeper manage their entire store through simple natural-language messages — no forms, no dashboards, no training required.

The bot is built on **Google Gemini's function-calling API** (google-genai SDK), where the AI decides which tool to invoke based on what the user says. All business data is stored in a **PostgreSQL database (Neon)** hosted on the cloud, and the bot runs 24/7 on **Render.com** using webhook mode.

What the bot can do:

- Receive new stock into the inventory
- Create multi-item bills with GST calculation (CGST + SGST, Decimal-accurate)
- Edit a bill draft before finalizing (change qty, remove items)
- Finalize a bill — atomically deducts stock, prevents double-finalization
- Block overselling — bot refuses if stock is insufficient
- Manage Khata (credit ledger) — add credit, record payments, show balance
- Generate GST-correct PDF invoices with line-item breakdown
- Generate PowerPoint analysis decks (sales, top items, stock health, GST charts)
- Remember owner preferences across sessions (e.g., store name, GST number)
- Low-stock alerts and reorder suggestions

■ ■ Technology Stack

Layer	Technology	Purpose
AI / Agent	Google Gemini (google-genai SDK)	LLM reasoning + function-calling
Bot Interface	python-telegram-bot (webhook mode)	User communication via Telegram
Database	PostgreSQL on Neon (free tier)	Persistent storage of all data
ORM	SQLAlchemy	DB models, migrations, queries

PDF	ReportLab	GST invoice generation
PPTX	python-pptx + matplotlib	Analysis deck generation
Hosting	Render.com (Docker, free tier)	24/7 deployment with auto-deploy
Language	Python 3.11	Core application
Testing	pytest (30 tests)	Unit + hardening tests

■ Deliverables Checklist

■	Live Bot	@Ops_agent_supermarket_bot — running on Render, accessible via Telegram app or Telegram Web
■	Agent Harness	google-genai function-calling SDK (equivalent to Claude Agent SDK / Vercel AI SDK)
■	Skills & Tools	21 LLM-callable tools covering billing, inventory, khata, analytics, PDF, PPTX, preferences
■	PDF Invoices	GST-correct ReportLab PDF — generated on request by the agent
■	PPTX Analysis Deck	python-pptx + matplotlib deck with real sales, stock, GST charts
■	README	Harness rationale, control loop, tool design, hard-requirement solutions — on GitHub
■	4–5 min Recording	Record using Telegram Web — demo script provided below
■	GitHub Collaborators	Invite Aswath363, akshaiP, ashwanthnebula (manual step — see instructions below)

■ Demo Recording Script (4–5 minutes)

Open Telegram Web → https://web.telegram.org/k/#@Ops_agent_supermarket_bot and record your screen while following these steps:

#	Scenario	Message to Send
1	Receive Stock	"50 packets of Maggi came in, cost ■12, MRP ■14"
2	Multi-item Bill	"Bill 5 packets of Maggi and 2 Tata Salt"
3	Edit Bill	"Make it 3 packets of Tata Salt instead of 2"
4	Finalize Bill	"Finalize bill with cash"
5	Oversell Guard	"Customer wants 500 packets of Maggi" → blocks it
6	Khata Credit	"Add ■500 to Ramesh's khata"

7	Khata Payment	"Ramesh paid ₹200 towards his account"
8	Khata Balance	"What is Ramesh's balance?" → ₹500 - ₹200 = ₹300
9	PDF Invoice	"Generate a PDF invoice for the last bill"
10	PPTX Deck	"Generate a sales analysis presentation deck"
11	Set Preference	"Set store name to Sri Lakshmi Supermarket"
12	Memory After /new	Send /new → "What is my store name?" → remembered ₹

■ ■ Tool Surface (21 Tools)

Category	Tools
Inventory	search_product, add_product, receive_stock, get_stock, get_low_stock, list_all_products
Billing	create_bill_draft, add_bill_item, update_bill_item, remove_bill_item, get_bill_draft, finalize_bill
Khata	create_customer, search_customer, add_credit, record_payment, get_customer_balance
Analytics	get_daily_sales, get_gst_collected
Documents	generate_invoice_pdf, generate_sales_analysis_pptx
Preferences	set_preference, get_preference

■ Hard Requirements & How They're Solved

Requirement	How It's Solved
Oversell Guard	Atomic SQL: UPDATE ... WHERE qty >= :qty — rollback if rowcount=0
GST Correctness	Decimal arithmetic, ROUND_HALF_UP per line, 50/50 CGST/SGST split
Multi-turn Drafts	Draft items stored in DB; stock untouched until finalize()
Idempotency	finalize() checks idempotency_key — duplicate calls return existing bill
Concurrency	Atomic conditional UPDATE (not SELECT-then-UPDATE); WAL on SQLite
Grounding	System prompt forbids invented numbers; all figures queried from DB
Guardrails	BelowCostError in finalize(); no delete_stock tool; payment guards balance
Real Artifacts	ReportLab PDF with line-item table; python-pptx+matplotlib with real data
Durable Memory	owner_preferences table in Postgres; get_preference called every turn
Direct Profit	direct_profit = Revenue – COGS (qty × cost_price), returned in JSON
Khata Balance	outstanding_balance = total_credit – total_payment, returned as JSON